Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If A is an orthogonal matrix of order 3 and $\mathrm{B} = \begin{bmatrix} 1 & 2 & 3 \\ -3 & 0 & 2 \\ 2 & 5 & 0 \end{bmatrix}$, then which of the following is/are correct?

1 $|\mathrm{AB}| = \pm 47$

2 $\mathrm{AB} = \mathrm{BA}$

Select the correct answer using the code given below:

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2

Q.2

If $[a]$ denotes the integral part of a and $x = a_3 y + a_2 z$, $\mathrm{y} = \mathrm{a_1 z} + \mathrm{a_3 z}$ and $\mathrm{z} = \mathrm{a_2 x} + \mathrm{a_1 y}$, where $\mathrm{x, y, z}$ are not
all zero. If $a_1 = m - [m]$, m being a non-integral constant, then $\mathrm{a_1 a_2 a_3}$ is

(A) > 1

(B) > -1

(C) < 1

(D) < -1

Q.3

The determinant $\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$ is independent of

(A) x only

(B) θ only

(C) x and θ both

(D) None of these

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Q.4

If $f(x)=\begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$ then $\lim_{x\to 0}\left[\frac{f'(x)}{x}\right]$ is

(A) 2

(B) -2

(C) 1

(D) -1

Q.5

In a third order determinant, each element of the first column consists of sum of two terms, each element of the second column consists of sum of three terms and each element of the third column consists of sum of four terms. Then it can be decomposed into n determinants, where n has the value

(A) 1

(B) 9

(C) 16

(D) 24

Q.6

If A is a square matrix of order n, then $\text{adj}(\text{adj } A)$ is equal to

(A) $|A|^{n-1}\ A$

(B) $|A|^{n}A$

(C) $|A|^{n-2}\ A$

(D) None of these

Q.7

Matrix M_r is defined as $M_r=\begin{pmatrix} r & r-1 \\ r-1 & r \end{pmatrix}$, $r\in N$. The value of $\det(M_1)+\det(M_2)+\det(M_3)+\ldots.+\det(M_{2014})$ is

(A) 2013

(B) 2014

(C) $(2013)^2$

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(D) $(2014)^2$

Q.8

If $a \neq b \neq c$ are all positive, then the value of the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is

(A) non-negative

(B) non-positive

(C) negative

(D) positive

Q.9

If $\mathrm{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ is $a 2 \times 2$ matrix and $\mathrm{f}(\mathrm{x}) = \mathrm{x}^2 - \mathrm{x} + 2$ is a polynomial, then what is $\mathrm{f}(\mathrm{A})$?

(A) $\begin{bmatrix} 1 & 7 \\ 1 & 7 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & 6 \\ 0 & 8 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 6 \\ 0 & 6 \end{bmatrix}$

(D) $\begin{bmatrix} 2 & 6 \\ 0 & 7 \end{bmatrix}$

Q.10

The maximum and minimum value of (3×3) determinant whose elements belongs to $\{0, 1\}$ is

(A) $1, -1$

(B) $2, -2$

(C) $4, -4$

(D) None of these

Q.11

If $A = \begin{matrix} 1 & \log_b a \\ \log_a b & 1 \end{matrix}$, then $|A|$ is equal to

(A) 1

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(B) 0

(C) $\log_a b$

(D) $\log_b a$

Q.12

If $a,\quad b,\quad c$ are distinct and rational numbers then $\begin{matrix} a^2+b^2+c^2 & ab+bc+ca & ab+bc+ca \\ ab+bc+ca & a^2+b^2+c^2 & ab+bc+ca \\ ab+bc+ca & ab+bc+ca & a^2+b^2+c^2 \end{matrix}$ is always

(A) zero

(B) Rational & Positive

(C) Rational & Negative

(D) Irrational and Positive

Q.13

If A is a square matrix such that $\mathrm{A}^2 = \mathrm{I}$ where I is the identity matrix, then what is A^{-1} equal to?

(A) $\mathrm{A}+1$

(B) Null matrix

(C) A

(D) Transpose of A

Q.14

If $\operatorname{adj} \mathrm{B} = \mathrm{A}, |\mathrm{P}| = |\mathrm{Q}| = 1$, then $\operatorname{adj}\left(\mathrm{Q}^{-1}\,\mathrm{BP}^{-1}\right)$ is

(A) PQ

(B) QAP

(C) PAQ

(D) $\mathrm{PA}^{-1}\mathrm{Q}$

Q.15

Let $\mathrm{A} = [\mathrm{a_{ij}}]_{\mathrm{m}\times\mathrm{m}}$ be a matrix and $\mathrm{C} = [\mathrm{c_{ij}}]_{\mathrm{m}\times\mathrm{m}}$
be another matrix where c_{ij} is the cofactor of a_{ij} Then, what is the value of $|\mathrm{AC}|$?

(A) $|\mathrm{A}|^{\mathrm{m}-1}$

(B) $|\mathrm{A}|^{\mathrm{m}}$

(C) $|\mathrm{A}|^{\mathrm{m}+1}$

(D) Zero

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Q.1 Answer:

Both 1 and 2

Solution:

The determinent of a orthogonal matrix is always ± 1 $|\mathrm{A}| = \pm 1$

$$\mathrm{B} = \begin{bmatrix} 1 & 2 & 3 \\ -3 & 0 & 2 \\ 2 & 5 & 0 \end{bmatrix}$$

$$|\mathrm{B}| = -10 - 2(-4) + 3(-15)$$
$$= -47$$
$$|\mathrm{AB}| = |\mathrm{A}||\mathrm{B}|$$
$$= (\pm 1)(-47)$$
$$= \pm 47$$

Taking A as identity matrix we can prove $\mathrm{AB} = \mathrm{BA}$

Q.2 Answer:

> -1

Solution:

Given, $x = a_3 y + a_2 z$

$\ldots$

$\mathrm{y} = \mathrm{a}_1\mathrm{z} + \mathrm{a}_3\mathrm{x}$

$\mathrm{z} = \mathrm{a}_2\mathrm{x} + \mathrm{a}_1\mathrm{y}$

Since, x, y, z **are not all zero, therefore given system of equations has non-trivial solution.**

$$\therefore \begin{vmatrix} 1 & -a_3 & -a_2 \\ a_3 & -1 & a_1 \\ a_2 & a_1 & -1 \end{vmatrix} = 0$$

$\Rightarrow \mathrm{a}_1^2 + \mathrm{a}_2^2 + \mathrm{a}_3^2 + 2\mathrm{a}_1\mathrm{a}_2\mathrm{a}_3 = 1 \quad \ldots$

Since, $a_1 = m - [m]$ **and** m **is not an integer.** $\therefore 0 < \mathrm{a}_1 < 1 \Rightarrow 0 < 1 - \mathrm{a}_1^2 < 1$

From Eq. (iv), $1 - a_2^2 - a_3^2 = a_1^2 + 2a_1a_2a_3$

$\Rightarrow 1 - a_2^2 - a_3^2 + a_2^2a_3^2 = a_1^2 + 2a_1a_2a_3 + a_2^2a_3^2$

$\Rightarrow \left(1 - a_2^2\right)\left(1 - a_3^2\right) = \left(a_1 + a_2a_3\right)^2$

Similarly, $\left(1 - a_1^2\right)\left(1 - a_3^2\right) = \left(a_2 + a_1a_3\right)^2 \ldots$ **(vii)**

$\left(1 - a_1^2\right)\left(1 - a_2^2\right) = \left(a_3 + a_1a_2\right)^2$

From Eq. (viii), $1 - \mathrm{a}_2^2 => 0 \cdot \frac{\left(a_3 + a_1a_2\right)^2}{1 - a_1^2}$

From Eq. (viii), $1 - \mathrm{a}_3^2 > 0 \Rightarrow 3 - \left(\mathrm{a}_1^2 + \mathrm{a}_2^2 + \mathrm{a}_3^2\right) > 0$

$\Rightarrow a_1^2 + a_2^2 + a_3^2 < 3 \Rightarrow 1 - 2a_1a_2a_3 < 3$

[From Eq. (iv)] $\Rightarrow \mathrm{a}_1\mathrm{a}_2\mathrm{a}_3 > -1$

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Q.3 Answer:

θ only

Solution:

$$\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$$

$$= x\left(x^2-1\right) - \sin\theta(-x\sin\theta - \cos\theta)$$
$$+ \cos\theta(-\sin\theta + x\cos\theta)$$
$$= -x^3 - x + x\sin^2\theta + \sin\theta\cos\theta$$
$$- \cos\theta\sin\theta + x\cos^2\theta)$$
$$= x^3 - x + x = x^3$$

Q.4 Answer:

-2

Solution:

We have, $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix} = \begin{vmatrix} \cos x - \tan x & 0 & 0 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$

[Applying $\mathrm{R}_1 \to \mathrm{R}_1 - \mathrm{R}_3$**]**

$= (\cos x - \tan x)\left(x^2 - 2x^2\right)$ [**Expanding along** R_1]

$= -x^2(\cos x - \tan x)$

$\therefore f'(x) = -2x(\cos x - \tan x) - x^2\left(-\sin x - \sec^2 x\right)$

$\therefore \lim_{x\to 0}\left[\frac{f'(x)}{x}\right] = \lim_{x\to 0} -2(\cos x - \tan x)$

$+x\left(\sin x + \sec^2 x\right)$

$2 \times 1 = -2$

Q.5 Answer:

24

Solution:

Column one can be divided into 2 determinants.

In each of those two determinants, column 2 can be divided into 3 different determinants, thus making a total of 6 . And each of those 6 determinants can be divided into 4 determinants, by seperating out srd column.

Hence, total determinants thus formed $= 2 \times 3 \times 4 = 24$

Q.6 Answer:

$|\mathrm{A}|^{\mathrm{n}-2}\ \mathrm{A}$

Solution:

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For any square matrix X, we have $\mathrm{X}(\operatorname{adj} \mathrm{X})=|\mathrm{X}| \mathrm{I}_{\mathrm{n}}$

Taking $\mathrm{X}=\operatorname{adj} \mathrm{A}$, **we get** $(\operatorname{adj} \mathrm{A})[\operatorname{adj}(\operatorname{adj} \mathrm{A})]=|\operatorname{adj} \mathrm{A}| \mathrm{I}_{\mathrm{n}}$

$\Rightarrow(\operatorname{adj} \mathrm{A})[\operatorname{adj}(\operatorname{adj} \mathrm{A})]=|\mathrm{A}|^{\mathrm{n}-1} \mathrm{I}_{\mathrm{n}}$

$\left[\because|\operatorname{adj} \mathrm{A}|=|\mathrm{A}|^{\mathrm{n}-1}\right]$

$\Rightarrow(\mathrm{A} \operatorname{adj} \mathrm{A})[\operatorname{adj}(\operatorname{adj} \mathrm{A})]=|\mathrm{A}|^{\mathrm{n}-1} \mathrm{~A}$

$[\because \mathrm{A}\ I_n=A]$

$\left(|\mathrm{A}| \mathrm{I}_{\mathrm{n}}\right)(\operatorname{adj}(\operatorname{adj} \mathrm{A}))=|\mathrm{A}|^{\mathrm{n}-1} \mathrm{~A}$

$\Rightarrow \operatorname{adj}(\operatorname{adj} A)=|A|^{n-2} A$

Q.7 Answer:

$(2014)^2$

Solution:

$$\det(\mathrm{Mr})=\begin{vmatrix} \mathrm{r} & \mathrm{r}-1 \\ \mathrm{r}-1 & \mathrm{r} \end{vmatrix}=2\mathrm{r}-1$$

$\sum_{r=1}^{2014} \det\left(M_r\right)=2 \sum_{r=1}^{2014} r-2014$

$=2 \times \frac{2014 \times 2015}{2}-2014=(2014)^2$

Q.8 Answer:

negative

Solution:

$$\begin{vmatrix} \mathrm{a} & \mathrm{b} & \mathrm{c} \\ \mathrm{b} & \mathrm{c} & \mathrm{a} \\ \mathrm{c} & \mathrm{a} & \mathrm{b} \end{vmatrix}=\begin{vmatrix} \mathrm{a}+\mathrm{b}+\mathrm{c} & \mathrm{b} & \mathrm{c} \\ \mathrm{a}+\mathrm{b}+\mathrm{c} & \mathrm{c} & \mathrm{a} \\ \mathrm{a}+\mathrm{b}+\mathrm{c} & \mathrm{a} & \mathrm{b} \end{vmatrix}$$

$\left(\because \mathrm{C}_1 \rightarrow \mathrm{C}_1+\mathrm{C}_2+\mathrm{C}_3\right)$

$$=(a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$$

[on taking $(\mathrm{a}+\mathrm{b}+\mathrm{c})$ **common from** $\mathrm{C}_1]$

$=(a+b+c)\left[1\left(bc-a^2\right)-b(b-a)+c(a-c)\right.$

$=(a+b+c)\left[bc-a^2-b^2+ab+ac-c^2\right]$

$=(a+b+c)\left[-\left(a^2+b^2+c^2-ab-bc-ca\right)\right]$

$=-\frac{1}{2}(a+b+c)\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$

Hence, the determinant is negative value

Q.9 Answer:

$$\begin{bmatrix} 2 & 6 \\ 0 & 8 \end{bmatrix}$$

Solution:

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Given that, $\mathrm{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

$\mathrm{A}^2 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2+6 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}$

Since, $f(x) = x^2 - x + 2$

Putting A in place of x $\mathrm{f(A)} = \mathrm{A}^2 - \mathrm{A} + 2\mathrm{I}$

$= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$= \begin{bmatrix} 1-1+2 & 8-2+0 \\ 0-0+0 & 9-3+2 \end{bmatrix}$

$\begin{bmatrix} 2 & 6 \\ 0 & 8 \end{bmatrix}$

Q.10 Answer:

$2, -2$

Solution:

(b) $|\mathrm{A}|_{\max} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = 0 - (-1) + 1(1) = 2$

$\therefore |\mathrm{A}|_{\min} = -2$

Q.11 Answer:

0

Solution:

We have, $A = \begin{matrix} 1 & \log_b a \\ \log_a b & 1 \end{matrix}$

$\therefore A = 1 - \log_a b \log_b a = 1 - 1 = 0$

Q.12 Answer:

Rational & Positive

Solution:

$\begin{matrix} a & b & c \\ b & c & a \\ c & a & b \end{matrix} \begin{matrix} a & b & c \\ b & c & a \\ c & a & b \end{matrix}$

$= \begin{matrix} a & b & c \\ b & c & a \\ c & a & b \end{matrix}^2$ **= Rational & Positive**

Q.13 Answer:

A

Solution:

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$A \rightarrow$ **square matrix** $A^2 = I$

To find $A^{-1} = ?$

Given $A^2 = I$

$\Rightarrow A \times A = I$

$\Rightarrow A = \frac{I}{A}$

$\Rightarrow A = A^{-1}$

$\therefore A^{-1} = A$

Option C is correct

Q.14 Answer:

PAQ

Solution:

Given adj $B = A, |P| = |Q| = 1$

Consider , adj $\left(Q^{-1}BP^{-1}\right) = \left(\text{adj } P^{-1}\right)\left(\text{adj } B\right)\left(\text{adj } Q^{-1}\right)$

$= (\text{adj } P)^{-1} \text{ A.} (\text{adj } Q)^{-1} = \left(P^{-1}\right)^{-1} \text{ A } \left(Q^{-1}\right)^{-1}$

$= PAQ$

Q.15 Answer:

$|A|^{m+1}$

Solution:

Let $A = [a_{ij}]_{m \times m}$ **be a matrix and** $C = [c_{ij}]_{m \times m}$ **be another matrix where** c_{ij} **is the cofactor of** a_{ij} **.** $\therefore$ **The value of** $|AC| = |A|^{m+1}$

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